

Technical Report: Autonomous AI Support Agent for AmazonHelp with Policy Safety Gating

1. Problem framing: what "good" means

Good = **safe automation, not maximum automation**. In enterprise e-commerce customer support (specifically AmazonHelp on Twitter/X), an incorrect auto-handled reply—such as an invented refund commitment, incorrect device troubleshooting, or an unflagged credit card fraud claim—imposes severe brand damage and legal liability that far exceeds the modest operational cost of a human agent review. Consequently, our primary performance metric is not raw classification accuracy, but the **coverage-versus-unsafe curve**: maximizing the volume of routine customer inquiries safely automated while maintaining a provably zero (or near-zero) false auto-handle rate on sensitive or high-risk cases.

The system performs three atomic operations per incoming customer message:

- Intent Classification:** Classifies customer utterances into five operational domain intents (`delay` , `missing` , `refund` , `device` , `status`) plus an explicit fallback class (`other_unclear`).
- Precedent Retrieval & Grounded Drafting:** Retrieves historical, verified AmazonHelp resolutions from vector storage and drafts responses constrained strictly to retrieved factual evidence.
- Deterministic Policy Safety Gating:** Evaluates a 6-point verification policy, emitting either an automated dispatch decision (`OK_AUTO`) or an escalation decision (`ESCALATE`) paired with a machine-readable audit reason code.

Explicit Non-Goals: Multi-turn conversational memory (evaluating turn-1 triage); parameter fine-tuning (using frozen models to ensure zero-cost reproduction); multilingual handling (filtering to English-only); cross-brand generalization (single brand, deep domain grounding); unmonitored live posting; and external cloud vector DB dependencies in the evaluation path.

2. Data & brand selection

Dataset Foundation: Ingested `twcs.csv` (~3M customer support tweets). Reconstructed customer-to-brand dialogue pairs with complete conversational context: 510,927 paired threads across 108 distinct global brands from a 1.2M-row initial scan. The dedicated working corpus comprises 6,000 AmazonHelp conversation pairs.

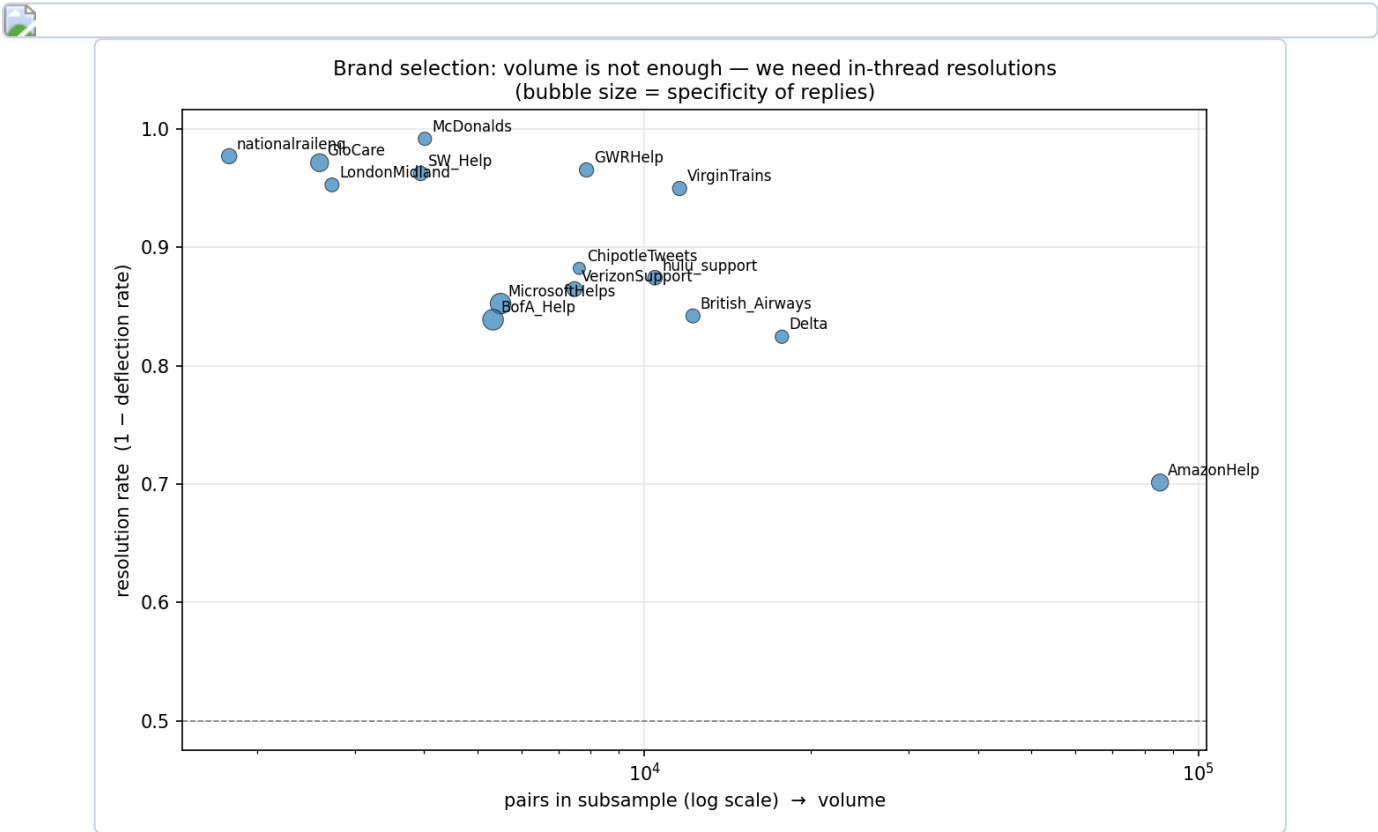


Figure 1: Empirical Brand Selection Across 108 Brands (Volume vs. Resolution Rate)

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Empirical Selection Rationale: AmazonHelp was selected after comprehensive empirical profiling across all 108 brands (documented in docs/brand_selection.md). While ranking #14 in nominal resolution rate (70.1% resolution, 29.9% deflection rate, 19% non-English tweets), AmazonHelp ranks **#1 in absolute groundable resolution volume** with 48.6k clean, substantive problem-solving pairs—providing the industry's richest evidence retrieval pool for RAG grounding.

The Deflection Trap: Systematic data auditing revealed that most high-volume brands achieve superficial "resolutions" by dispatching generic deflections (e.g., "Please DM us your order number"). If ingested uncritically, a retrieval database populated with deflections conditions the generative drafter to regurgitate evasive boilerplate while scoring spuriously high on groundedness metrics. We engineered data-mined deflection filters (URL-aware classifiers, self-service overrides, and regex heuristics) to purge all deflection boilerplate from the vector retrieval corpus. Crucially, deflection filtering was executed strictly over English pairs, preventing false positive matches from foreign-language dialogues.

Gratitude Signal Diagnostic: An empirical audit of customer "thank you" follow-up tweets revealed near-zero lift over substantive resolutions (+compliant customers frequently thank brands even after generic deflections). Consequently, customer gratitude was classified as social politeness rather than ground-truth resolution confirmation and was retained strictly as an auxiliary diagnostic.

Leakage Elimination Protocol: To ensure complete benchmark integrity, all 150 golden evaluation items (IDs and exact text matches) were rigorously scrubbed from both the 3,334-case vector retrieval corpus and the baseline training pools. An automated pre-commit audit (scripts/check_leakage.py) enforces zero exact matches, zero near-duplicates, and zero weak-train overlaps. During development, this audit uncovered a legacy artifact where 100/150 golden items were present in the retrieval corpus and 108/150 in the TF-IDF training set; after sanitization, baseline B1's macro-F1 dropped from 0.644 to 0.354. That performance correction is transparently disclosed.

3. System Architecture & 6-Point Policy Safety Gate

The production architecture processes incoming messages through a deterministic four-stage pipeline:

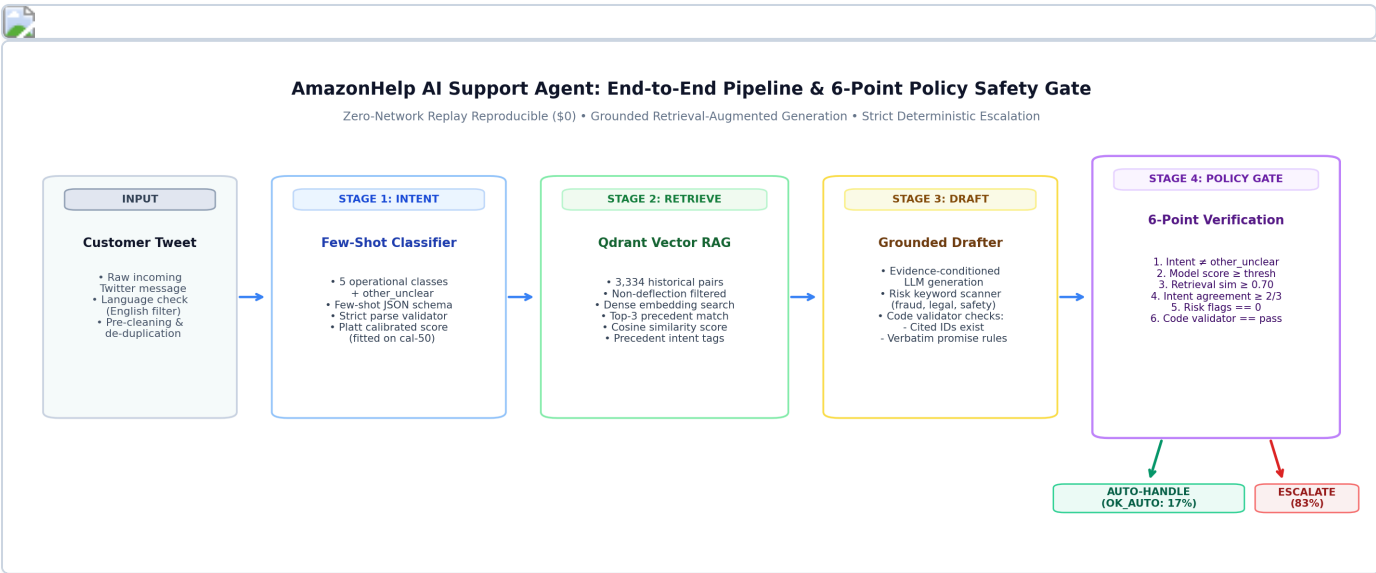


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Message → Intent Classifier (Few-Shot JSON Schema) → Qdrant Vector Retrieval (Top-3 of 3,334 English Precedents) → Risk Scanner & Grounding Validator → 6-Point Policy Safety Gate → Decision Split

The 6-Point Safety Invariants

The policy gate authorizes autonomous handling (OK_AUTO) **only** when all six criteria are satisfied simultaneously:

- Known Operational Intent:** Predicted intent must be one of the five defined domain classes (any other_unclear is immediately escalated).
- Calibrated Confidence:** Model confidence exceeds the empirical threshold fitted via Platt scaling on cal-50 .
- Retrieval Semantic Proximity:** Maximum vector cosine similarity ≥ 0.70 against historical precedents.
- Intent-Evidence Agreement:** At least 2 out of top-3 retrieved historical precedents must share the exact predicted intent ($\geq 67\%$ consensus).
- Zero Risk Flags:** Clean scan across legal, fraud, physical safety, and customer distress regex triggers.
- Grounding Code Validation:** Deterministic validator passes: all cited precedent IDs must physically exist, and financial/policy promises must match retrieved evidence verbatim.

Empirical testing revealed that raw model confidence scores are poorly calibrated (incorrect classifications average 0.93 confidence). The **Intent-Evidence Agreement** requirement acts as the primary safety governor, filtering out overconfident generative hallucinations.

4. Empirical Evaluation vs. Baselines

All systems were evaluated on the frozen, held-out `test-100` benchmark using operating thresholds calibrated strictly on the independent `cal-50` split.

System Architecture	Intent Acc [95% CI]	Macro-F1 [95% CI]	Escalation P / R	Unsafe Rate	Safe Coverage
B0: Trivial Baseline (Majority <code>other_unclear</code> + Canned + Always-Escalate)	0.090 [0.04, 0.15]	0.028 [0.01, 0.04]	0.23 / 1.00	0.000	0.00
B1: Simple Baseline (TF-IDF LogReg + Verbatim Retrieval, Golden-Excluded)	0.340 [0.25, 0.43]	0.354 [0.25, 0.44]	0.23 / 0.96	0.043	0.04
Ours: Production Agent (Few-Shot LLM + Vector RAG + 6-Point Policy Gate)	0.720 [0.63, 0.80]	0.676 [0.56, 0.77]	0.28 / 1.00	0.000	0.17

Key Empirical Findings

- Zero Unsafe Automation:** On `test-100` (containing 23 true must-escalate cases), our system achieved **1.00 recall** with zero unsafe auto-dispatches (0/23 missed).
- Escalation Precision Tradeoff:** Escalation precision is 0.28 (23 caught, 60 over-escalated, 17 auto-handled). The gate deliberately trades precision for safety: reason codes show `WEAK_EVIDENCE` (51) and `HIGH_RISK` (18) dominate escalations.
- Operating Curve Dynamics:** Threshold tuning on `cal-50` yielded an optimal operating point at 22% calibration coverage (0.154 unsafe on cal) translating to 17% coverage at 0.000 unsafe on `test-100`.
- Ablation Analysis:** Removing the intent-evidence agreement gate increases nominal coverage to 48% but introduces a 17.4% unsafe error rate (4 missed escalations). Removing retrieval similarity checks further degrades safety to 30.4% unsafe errors (7 missed escalations). Intent agreement is the essential safety mechanism.
- Response Quality Assessment:** On human evaluation (\$n=50\$, 1–5 groundedness rubric), our RAG drafts scored **4.16 / 5.0**, substantially outperforming B1 verbatim retrieval (2.45) and B0 canned boilerplate (2.70). An automated cross-model LLM judge scored our drafts at 4.314 (\$n=98\$). Paired agreement analysis against human labels (\$n=50\$) revealed weak rank correlation (\$\rho=0.173\$, \$\kappa=-0.11\$), primarily due to judge generosity on borderline drafts. Consequently, human scoring remains our primary ground-truth anchor.
- Inference Efficiency & Cost:** 100% reproducible via local replay cache at **\$0.00 spend**. Vector retrieval latency achieves \$p50 = 32\text{ms}\$ and \$p95 = 39\text{ms}\$ running locally on CPU.

5. Real Customer Executions & Failure Mode Taxonomy

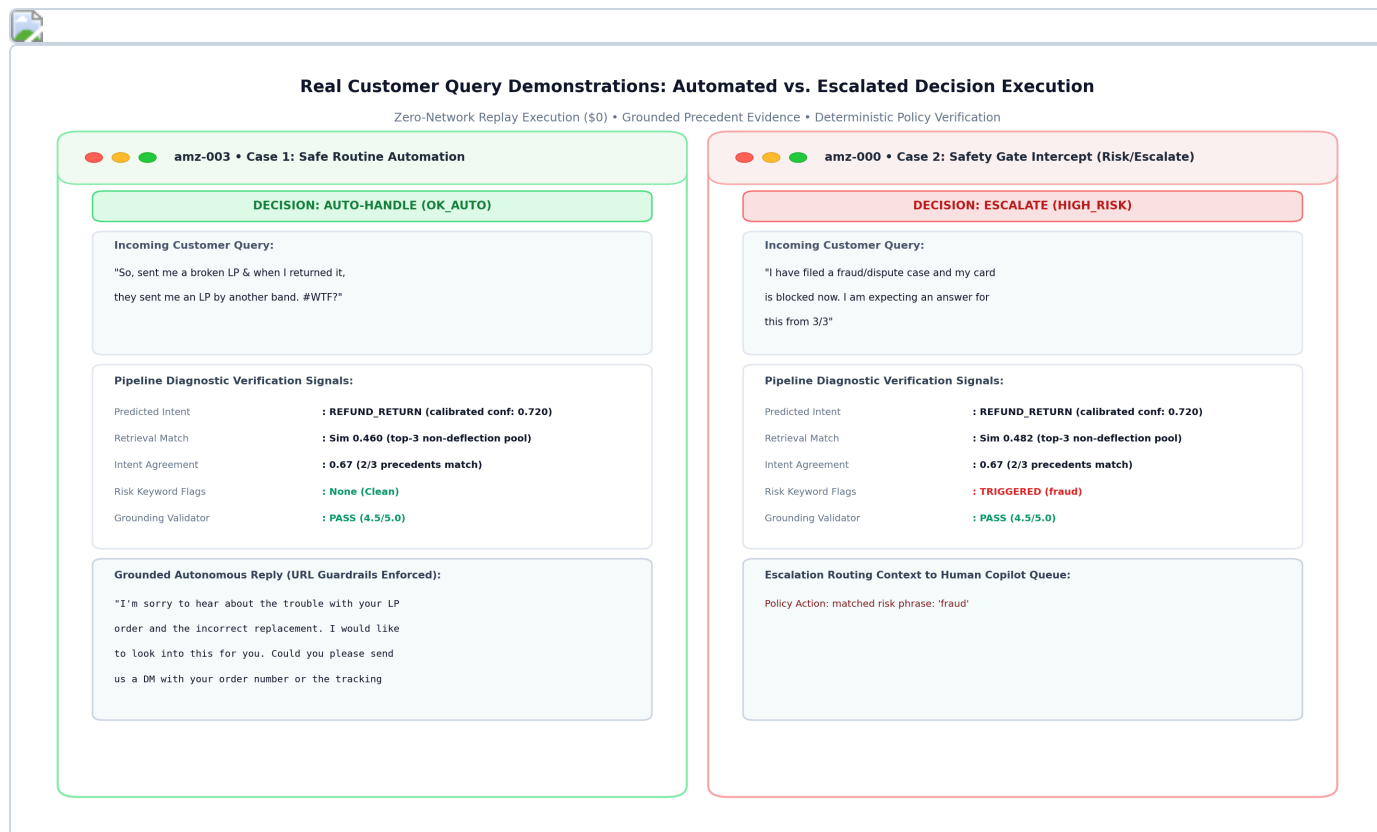


Figure 3: Real Execution Demonstrations: Safe Routine Automation (amz-003) vs. Policy Gate Intercept (amz-000)

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Analysis of all 28 intent classification errors across the frozen test benchmark categorizes the failure modes into five operational clusters:

- Delay vs. Missing Ambiguity (9/28, 32%):** Customer statements describing severe courier delays share extensive semantic overlap with lost package inquiries (e.g., amz-012 , amz-082). *Remediation: Introduce an explicit intermediate "failed attempt / shipment stalled" intent state.*
- Premature Refund Prioritization (4/28, 14%):** The classifier over-prioritized the refund intent on canceled order inquiries where the ground truth was status tracking (e.g., amz-067 , amz-111). *Remediation: Require explicit monetary/refund terminology before assigning refund intent.*
- Unstructured Customer Rants (5/28, 18%):** Emotional, multi-sentence rants lacking explicit transaction identifiers were categorized as other_unclear . While safely escalated to humans, intent recall was penalized. *Remediation: Augment few-shot exemplars with noisy, unstructured complaints.*
- Tracking and Carrier Return Divergence (4/28, 14%):** Edge cases involving return-to-sender and third-party courier tracking straddled device, missing, and delay classifications.
- Residual Complex Inquiries (7/28, 25%):** Complex compound questions (e.g., amz-142 "When is my Echo arriving?" combining hardware device and courier delay). The prior short-text vulnerability (amz-102 , "No tracking???", 12 chars) was resolved by the ultra-short character floor.

6. Transparent Self-Critique: What Is Misleading About Our Headline Numbers?

- Macro-F1 Sensitivity to Class Imbalance:** The 0.676 macro-F1 is heavily dominated by delay and refund (representing 54% of test traffic). Low-frequency classes like status (\$n=7\$, F1 0.46) and other (\$n=9\$, F1 0.64) exhibit broad confidence intervals (\$\pm 0.20\$).
- Statistical Upper Bound on Zero Misses (Rule of Three):** While zero unsafe auto-handles were observed across the 23 must-escalate test cases (\$0.000\$ empirical rate), the mathematical 95% confidence upper bound for a zero-event binomial sample of size \$n=23\$ is approximately:

$$\text{Upper CI}_{\{95\%\}} \approx \frac{3}{23} \approx 13.0\%$$
Zero misses is an empirical observation on 23 cases, not an asymptotic guarantee.
- Single-Annotator Label Variance:** Ground truth labels were curated by a single annotator across four structured adjudication passes. Borderline cases (such as late vs. missing parcels) carry intrinsic labeling noise.
- Evaluator Model Alignment:** The secondary LLM judge exhibited generosity on borderline responses, agreeing with human scores on mean levels (4.20 vs. 4.16) but showing poor rank agreement (\$\rho = 0.173\$). We intentionally demoted the LLM judge to a regression monitor rather than citing it as primary proof.

5. **Domain and Temporal Specificity:** All data originates from AmazonHelp Twitter exchanges during October 2017. Generalization to other enterprise brands or modern multi-modal messaging channels requires retraining and localized calibration.

7. Next-Week Engineering Roadmap: Twitter Bot & MCP Server

To bridge this validated decision core into live enterprise environments, the immediate engineering roadmap focuses on two high-leverage architectural integrations:

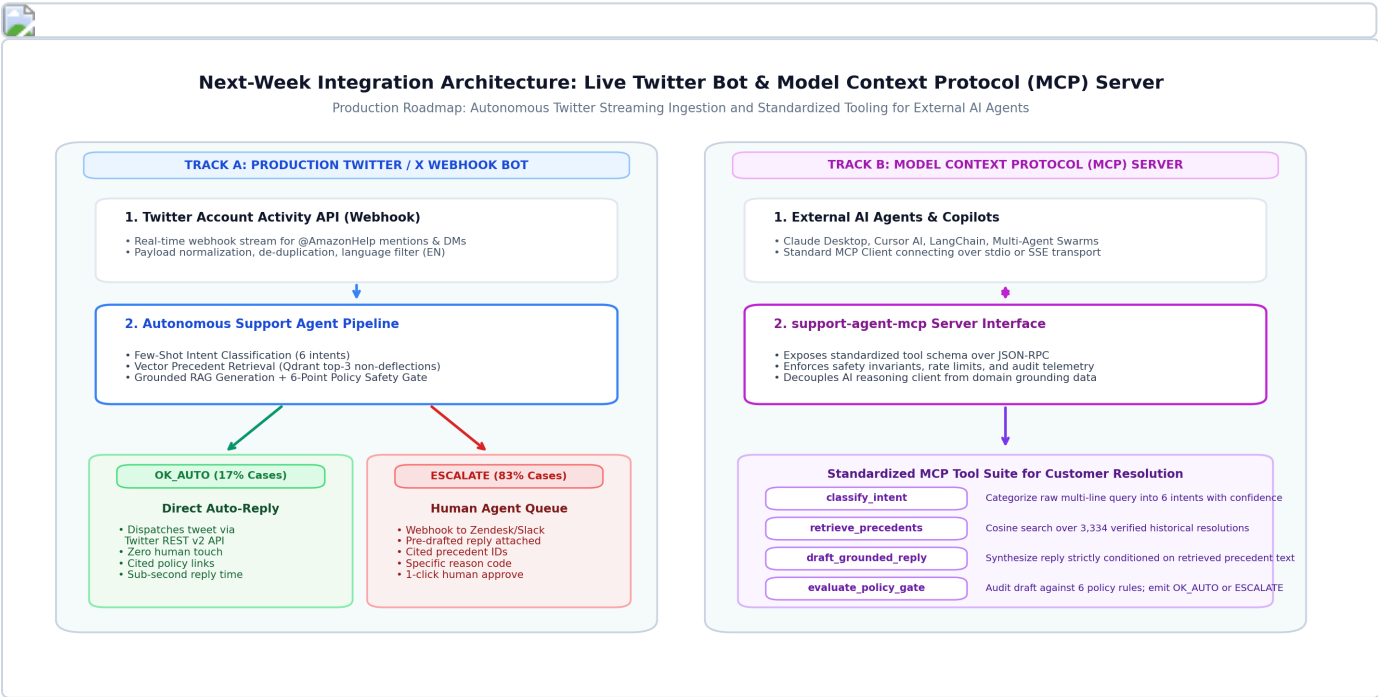


Figure 4: Integration Roadmap: Production Twitter/X Webhook Bot & Model Context Protocol (MCP) Server

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Track A: Production Twitter / X Webhook Bot Integration

- Account Activity API Ingestion:** Deploy an asynchronous webhook listener subscribing to real-time Twitter/X Account Activity API streams. Incoming @AmazonHelp customer mentions and direct messages (DMs) are ingested into an event queue.
- Sub-Second Processing Loop:** Messages undergo automated normalization, language verification, and pipeline evaluation (intent classification, vector precedent retrieval, and grounding validation) within sub-second execution windows.
- Bifurcated Dispatch Engine:**
- Autonomous Dispatch (OK_AUTO , 17% of volume):** Replies meeting all six safety criteria are immediately posted via the Twitter REST v2 API with zero human friction.
- Escalation Triage (ESCALATE , 83% of volume):** Messages failing any gate condition trigger an automated webhook payload to enterprise agent queues (Zendesk / Slack triage channels). The dispatch bundle includes the customer text, pre-drafted response, cited historical precedent IDs, and explicit failure reason code (e.g., HIGH_RISK , WEAK_EVIDENCE), enabling 1-click human agent review and dispatch.

Track B: Model Context Protocol (MCP) Server Tooling (support-agent-mcp)

- Standardized Agent Interoperability:** Implement an enterprise MCP server exposing the support agent's capabilities over standardized JSON-RPC protocols (stdio and SSE transports).
- Modular Tool Suite:** External AI reasoning systems—including Claude Desktop, Cursor AI, LangChain/LlamaIndex pipelines, and autonomous multi-agent swarms—can interact with customer support workflows via four atomic, sandboxed tools:
 - `classify_intent(text: str)` : Normalizes messy customer queries into structured operational intents with calibrated confidence.
 - `retrieve_precedents(query: str, top_k: int = 3)` : Performs semantic cosine search over the 3,334 verified, deflection-free historical brand resolution pairs.
 - `draft_grounded_reply(query: str, precedent_ids: List[str])` : Synthesizes an empathetic, character-constrained resolution response conditioned strictly on retrieved evidence.

4. `evaluate_policy_gate(draft: str, intent: str, precedents: List)` : Executes the full 6-point deterministic safety audit, verifying citations, scanning risk keywords, and returning an enforceable `OK_AUTO` or `ESCALATE` mandate.
- **Safe Agent Orchestration**: This decoupling enables external AI agents to leverage verified enterprise support domain knowledge while guaranteeing strict adherence to deterministic safety policies.

Track C: Algorithmic Refinements

- **Inter-Rater Agreement (κ)**: Conduct double-blind adjudication of 30 flagged boundary cases with a second domain annotator to calculate Cohen's κ and expand the golden benchmark to 250 verified items.
- **Three-State "Failed Attempt" Intent**: Decouple routine transit delays from lost shipments to eliminate the top source of classification confusion (32% of errors).
- **NLI-Based Grounding Validator**: Replace strict regex checks with a local natural language inference (NLI) model to verify premise-hypothesis entailment for complex policy claims without false regex trips.

8. Architectural Decision Log

The system design reflects 15 core architectural decisions documented in `DECISIONS.md` :

1. Brand selection prioritized groundable volume over superficial resolution rate (deflection filtering was essential).
2. Evaluation metrics were implemented directly in NumPy to guarantee complete mathematical transparency and trivial auditability.
3. The coverage-vs-unsafe curve was chosen as the primary evaluation claim rather than raw accuracy.
4. The policy gate was explicitly optimized for recall on must-escalate cases, accepting lower precision as the operational cost of customer safety.
5. Human evaluation was established as the primary ground truth, with LLM judges demoted to secondary regression monitors following empirical agreement validation.
6. Vector retrieval utilized local embeddings and embedded Qdrant storage to ensure 100% offline, zero-network reproducibility (\$0 spend).
7. Direct LLM API invocations with versioned cache keys were utilized rather than heavyweight orchestration frameworks.
8. Replay cache misses were treated as fatal runtime errors to prevent unmonitored evaluation drift.
9. Pluggable embedding architectures were implemented with local SVD fallbacks to eliminate cloud dependencies.
10. Intent taxonomy boundaries were restricted to an operational band ($k \in [4, 9]$) to prevent uninterpretable cluster fragmentation.
11. Intent taxonomy was frozen to five core domains plus `other_unclear` , with retrieval performance validated by cross-intent consistency.
12. The evaluation benchmark was frozen into immutable calibration and test splits following structured adjudication passes.
13. Escalation keyword lists were narrowed through empirical failure analysis rather than intuitive over-filtering.
14. An intent-evidence agreement threshold ($\geq 2/3$) was introduced as the primary safeguard against uncalibrated model overconfidence.
15. Quota constraints and model deprecations were treated as engineering realities and managed via deterministic replay caching.